

Melanoscope: Melanoma Detection System

Manya Bora
Department of Computer Science
Engineering
Maharaja Surajmal Institute of
Technology
(Affiliated to GGSIPU)
New Delhi, India
mbora31073@gmail.com

Avinash Thakur
Department of Computer Science
Engineering
Maharaja Surajmal Institute of
Technology
(Affiliated to GGSIPU)
New Delhi, India
thakuravinash409@gmail.com

Aditya Gupta
Department of Computer Science
Engineering
Maharaja Surajmal Institute of
Technology
(Affiliated to GGSIPU)
New Delhi, India
aditya13451@gmail.com

Abstract—Melanoma is a type of skin cancer that is caused in the melanin producing cells known as melanocytes. It can be fatal and requires fast action. It has high chances of arising in already existing moles so, people often disregard any shown symptoms until it has worsened. For effective treatment, it is beneficial to nip the cancer right in its bud. It is a highly invasive and aggressive types of skin cancer. Manual diagnosis via biopsy is very time consuming and might result in excision of a skin portion for naught. Therefore, we suggest a way to test the presence of melanoma by uploading skin lesion images to confirm any arising suspicion about an irregular mole or skin patch. This research employs the use of Convolutional Neural Networks to create a predictive analysis of melanoma through image segmentation. Two CNN architecture models are ensembled together to increase the performance accuracy. The HAM1000 dataset, collected in the ISIC challenge, is used to train and test the data. At first, due to a skewed ratio of the different classes described in the dataset, data augmentation is applied and oversampling is introduced. This plays a huge role in improving the robustness of the model. Multiple image segmentation techniques are then applied by rescaling the pixel values, randomly rotating the images, zooming images in and out by 10% and shifting images horizontally and vertically. Once the data is pre-processed, two CNN models are proposed with custom activation and loss functions and batch normalizations. Mean Squared Error (MSE) and Categorical Cross Entropy (CCE) are used as loss functions and Rectified Linear Unit (ReLU) and Leaky Rectified Linear Unit (Leaky ReLU) are used as activation functions. Both models attained an accuracy of 98 and 97 respectively. The aggregation of the two models is found to produce the best possible outcome for prediction. Each prediction is calculated separately and then, ensemble predictions are formed by merging the outputs of both model A predictions and model B predictions. Through this, a test accuracy of 98.5%, precision of 98.51, recall of 98.5% and an F1 score of 98.48% is achieved.

Keywords—melanoma, skin cancer, HAM1000, Deep learning, Convolutional Neural Network

I. INTRODUCTION

The skin is the largest organ of the human body. About 15% of a human's body weight is contributed by their skin. Since, a large part of our lives involves being exposed to the sun, skin has high chances of getting affected. Sunlight provides vitamin D which is essential for our body and therefore, cannot be avoided. This poses a huge problem as it also provides a gateway to UV light exposure. Although exposure to sunlight is beneficial to us, it is also one of the leading causes of melanoma and is found to accumulate the melanoma-causing factors over time.[1] Therefore, we must take precautions to ensure that any sign of this potentially life-threatening disease is diagnosed as early as possible. Along with this, distinct genetic alterations have been found to be connected with melanoma.[2]

Melanoma is also known as black tumour and is a type of skin cancer of the pigment-producing cells of the skin known as melanocytes. The lesions appear as irregularly shaped or unusually coloured marks on the body. [3] It is highly invasive and should be diagnosed in the early stages which allow successful treatment through surgery alone. Once the cancer cells start to spread to parts of the body other than the original tumour site, survival rates drop significantly.[4] To detect melanoma, a skin exam is conducted in which pigmented areas and existing moles are examined for abnormality.[5] Many people disregard the seriousness of skin conditions and avoid contacting the dermatologists. Therefore, we provide a way to identify melanoma through pictures. In this research, the concept of Convolutional Neural Networks is used to demonstrate a deep learning model that detects the presence or absence melanoma.

II. LITERATURE REVIEW

In [6], the authors proposed a deep-learning method for pixel-wise classification using softmax classifier. The system employed an enhanced encoder-decoder network connected through skip pathways. They attained an accuracy and dice coefficient of 95% and 92% on the ISIC dataset.

In [7], introduced a computer-aided detection technique for early melanoma diagnosis. They proposed two methods to investigate the relationship between model depth and performance with varying dataset sizes. The first method employs convolutional neural networks, AlexNet, LeNet, and VGG-16, while the second uses support vector machines with a default RBF kernel to arrive at an accuracy rate of 91% and 86.6% respectively.

In [8], Devika Moturi, Ravi Kishan Surapaneni and Ventaka Sai Geethika Avanigadda reiterate the importance of early detection and precise diagnosis for determining cancer therapies. The suggested technique utilises deep learning methods like MobileNetv2 and Dense net for classifying benign and malignant tumours.

In [9], the authors investigate the creation of an ensemble of deep convolutional neural networks to combine the strengths of different models and further improve the accuracy in identification of three classes of skin cancer. This approach allows to achieve great results even in the absence of a large dataset of labelled images. The outputs of four different deep neural networks, GoogleNet, AlexNet, ResNet, and VGGNet, are fused together to get average area under the receiver operating characteristic curve to be 0.891 on a dataset of the IEEE International Symposium on Biomedical Imaging (ISBI) 2017 challenge on Skin Lesion Analysis Towards Melanoma Detection.

In [10], squamous, basal cell and melanoma as the three primary subtypes of skin cancer are discussed. The authors

trained the model with HAM10000 dataset, a lesion-based image dataset. Enforcing an autoencoder for feature extraction followed by classification with the help of v3 pretrained model and 3D CNN leading to an accuracy of 0.96. The help of an autoencoder as a feature extraction brings enhances efficiency and performance.

In [11], aim to increase the precision and efficacy of skin cancer recognition and enhance patients' experiences. They integrate Standard U-Net and Improved MobileNet-V3 with optimization techniques in the deep learning training process which allows the system to differentiate between malignant and benign lesions. Several adjustments are made in the model architecture. A bias loss function is used in place of the crossed-entropy loss function in the MobileNet-V3 model. Optimization techniques such as the Bayesian method

In [12], two deep learning methods that focus on the three necessary tasks of image processing of skin lesions namely lesion segmentation, lesion feature extraction and lesion classification are introduced. The deep learning framework was experimented on the ISIC 2017 dataset by using a CNN model for feature extraction and two fully convolutional residual networks (FCRN) for segmentation result and classification. The classification results were refined by a lesion index calculation unit (LICU) and accuracies of 0.753, 0.848 and 0.912 were received for the three tasks respectively.

In [13], a merged system made with the combination of three different methods is proposed. The three methods include a convolutional neural network and two classical machine learning classifiers trained with a set of features describing the texture, colour and borders of the skin lesions. A colour enhancement technique is used to begin the preprocessing method. The 2-D derivatives of Gaussian (DOG) and thresholding is used for hair detection and separation, respectively. Some other techniques used for preprocessing are Otsu method, hierarchical clustering and morphological snakes. CNN along with KNN and SVM methods were each tested. Result aggregation of the different methods enhances the overall performance to an accuracy of 88.4%.

In [14], a deep learning based 'You Only Look Once (YOLO)' algorithm is proposed to detect melanoma from dermoscopic images. It works on the basis of the application of deep convolutional neural networks. They aim to offer a more precise alternative to conventional CNNs by incorporating a two-phase process combining graph theory and fuzzy number-based approximation heightened segmentation results. The concepts of minimal spanning tree and L-type fuzzy number were used and yielded a Jac score of 79.84% on the ISIC 2019 dataset.

In [15], an approach using deep learning as a method for the precise extraction of a lesion zone was put forward. To improve the image quality, Enhanced Super-Resolution Generative Adversarial Networks (ESRGAN) was used, followed by segmentation to segment Regions of interest (ROI) from the full image. The image is evaluated with a CNN and a modified version of Resnet-50 for classification of skin lesions. The CNN-based model gave an accuracy and precision of 0.86 and 0.84 respectively on the ISIC 2018 dataset.

In [16], digital image processing techniques along with Support Vector Machine is used for the detection of skin

cancer. It provides a step up from the biopsy method of skin cancer detection by eliminating the need for unnecessary excision of moles. Gray level co-occurrence matrix (GLCM) is used for feature extraction and captures features like contrast, mean, energy and homogeneity. SVM is then used for regression analysis and classification of cancerous and non-cancerous images using the features extracted by the GLCM as input.

In [17], training and testing of 11 convolutional neural network architecture models on the HAM10000 dataset, which include a variety of seven skin lesion classes, is provided. The data is fine-tuned and augmented to solve the imbalance problem for training before the testing is done. Through testing, DenseNet169 is found to produce the best results an outperforming existing state-of-the-art method. An accuracy of 92.25%, a recall of 93.59% and an F1-score of 93.27% is attained by the DenseNet169. An android mobile application was also constructed using a light version of the model to classify benign or malignant lesion types along with providing the user with information about the allowed sun exposure time based on the current UV radiation degree etcetera.

In [18], an Inception V2 network inspired convolutional neural network is employed to study the impact of image quality on the accuracy of diagnosis by sonification. A basic skin magnifier with polarized light (SMP). A k-means clustering algorithm was used to cluster the activations of which the one with the lowest error was chosen. A secondary approach using another deep learning model was also applied. SMP achieved a F2 score sensitivity of 91.7% while the DL approach received a 89.5%.

In [19], a method where the region of interest within a dermoscopic image is automatically cropped using the Mask and Region-based CNN technique is proposed. This is then followed by classification of lesion images as benign or malignant using a ResNet152 structure. It was evaluated on the ISIC 2017 challenge and suggests a better performance over the previously proposed methods. The eVida M6 model provides an overall accuracy of 0.904, sensitivity of 0.820 and specificity of 0.925.

In [20], a deep-learning based approach for the segmentation of melanoma images at its earliest stage is presented. Region-based Convolutional Neural Network (RCNN) is employed along with fuzzy k-means clustering (FKM). The dataset images are first pre-processed to enhance the characteristics, before applying the RCNN to obtain the feature vector of fixed length. The melanoma-affected portion is then segmented using FKM. The proposed method achieved an average accuracy of 95.49, 93.1 and 95.6% on the ISIC-2016, ISIC-2017, and PH2 datasets, respectively.

III. SCOPE OF WORK

The scope of this research includes the development of a predictive model to detect melanoma using high-definition images of skin lesions. The final model is made by aggregation of two different convolution neural network models with an objective to improve the accuracy and precision of the predictive system. The concept of deep learning is employed.

Data collection and pre-processing techniques like data augmentation, noise removal, image pixel resizing,

normalization and spatial shifts are incorporated to refine the dataset and set up an efficient foundation for training and testing of the deep learning models.

This project leverages established approaches from existing literature to craft a system that both, validates and improves upon it. The insights gained through the in-depth evaluation of multiple methodologies and frameworks, allow to build upon the former studies to optimize and enhance their performances. The high potential of deep learning in medical field allows contribution to the field of dermatology.

IV. MATERIALS AND METHODS

The methods and techniques involved in the development process of this system are described in this section.

A. Data Collection

HAM10000 is a dataset with a collection of over 10,000 skin lesion images comprising of melanocytic nevi, melanoma, benign keratosis-like lesions, basal cell carcinoma, actinic keratoses, vascular lesions, and dermatofibroma dermatoscopic categories. It was collected for the ISIC (International Skin Imaging Collaboration Laboratory) challenge and tackles the hurdles caused by the small sized data and lack of diversity available in training of neural networks for melanoma-related projects by gathering images from different populations.

B. Exploratory Data Analysis

To work with data, we need to understand it and any underlying relationships between the factors. The data is read from the csv file and using python libraries like numpy, pandas and seaborn, data is represented in a graphical format. Histograms, bar plots, box plots, correlation heatmaps, violin plot etc. are used for data visualisation thus, enhancing comprehension. Various classes of skin cancer along with factors like frequency, age distribution, gender, localisation are studied. A correlation heatmap of age vs gender, fig. 1, reveals that there is no significant correlation between age and gender as -0.17 is very close to 0.

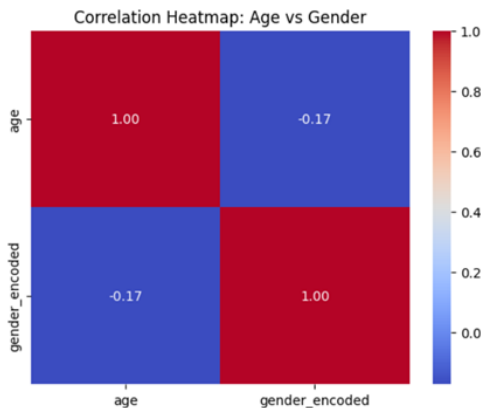


Fig. 1. Correlation heatmap of age vs gender

C. Data Pre-processing

The foundation of a machine learning model is a fine-tuned dataset with minimised errors so, we must clean the

dataset for any irregularities or bias occurring. By observing its shape, it can be inferred that the dataset is skewed towards a particular class. This might make the machine learning model biased towards the majority class and hence, making faulty predictions that favour one more than the others. This is prevented by introducing oversampling to the other classes and reaching a balance to acquire a bias-free collection. This is done by increasing the class samples by randomly duplicating certain examples from the minority class and adding them to it. The data is then reshaped to a 4D vector, (46935, 28, 28, 3), to better represent the image dataset by defining the number of samples now increased from 10,000 to 46,935, pixel size and the number of channels for RGB value respectively.

To make the dataset more suitable for training, the pixel size of each image is reduced along with application of random rotations by ten degrees and random zoom ins and outs by 10%. This makes the model more robust by allowing it access to different perspective views. The data is then split into a ratio of 80 to 20 for the training and testing datasets.

D. CNN

Convolutional Neural Networks (CNNs) are a type of deep neural network used to solve difficult image-driven pattern recognition and computer vision tasks [21]. It uses the concept of hierarchical feature detectors comprising of multiple layers such as convolutional layer, non-linearity layer, pooling layer and fully-connected layer that work in a way to recreate the functioning of a human brain [22]. Feature extraction is performed by the convolution and pooling layer which then allows the fully connected layer to maps the extracted features to final output [23]. A convolution layer contains a matrix (small grid) of parameters called kernel, which performs feature extraction, and a stack of mathematical operations which are applied at each layer, passing its output onto the next [23]. For our research, two CNN models are made and then, ensemble together to achieve best possible results. Dropout layer assigns each unit a fixed probability, independent of others, and temporarily removes certain units forward to prevent overfitting. [24] Batch Normalisation, a technique to normalise the activations in the intermediate layers of the network, is also applied to increase training speed and accuracy.[25]

E. Model Training

The two models, model A and model B, are trained and tested separately before creating the ensemble of the combination of the two. Below is given the training and test epoch loss and accuracy of models A and B using 10 epochs and a batch size of 128 units.

Both the models, model A and model B, have a similar architecture except a few features which are given below:

1) *Loss Function*: It measures the difference between the predicted output and the expected value and serves to minimise it.[26]

a) *Categorical Cross Entropy*: It is used in model A and is the measure of the difference between the probability distributions of the predicted class and the actual class labels. [27] Its formula is given below in eq 1.

$$CCE = - \sum_{i=1}^C y_i \log(\hat{y}_i) \quad (1)$$

b) *Mean Squared Error*: It is also known as L2 loss. In context to our research, it is used in model B and calculates the square of the error difference between the predicted and target image as seen in eq. 2.[28]

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (2)$$

2) *Activation function*: They are functions used to transform an input signal to an output signal which is then passed as input to the next neural network layer. They help make the network dynamic and allow the extraction of complex features to represent non-linear relationships between the input and output.[29]

a) *ReLU*: It stands for Rectified Linear Unit and is a mathematical function that assigns zero to values below zero. The output is otherwise given x if the input value is $x > 0$ as given in eq 3. Model A uses ReLU for activations.

$$f(x) = \max(0, x) \quad (3)$$

b) *Leaky ReLU*: Model B uses this as activation function. It is a mathematical function that solves the dying ReLU problem and reduces the possibility of having 0 as output by introducing a small constant ' α ' in cases of negative value to prevent zero outputs when there is a huge negative bias as given in eq. 4.[29]

$$f(x) = \begin{cases} x & \text{if } x > 0, \\ \alpha x & \text{if } x \leq 0 \end{cases} \quad (4)$$

3) *Weight Initialization*: Model A uses a normal initializer while model B uses a truncated normal initializer. Epoch vs loss accuracy curve is given in fig. 2 & 3.

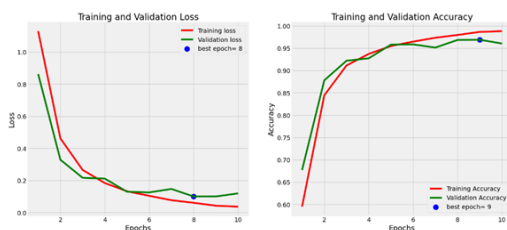


Fig. 2. Model A epoch vs loss/accuracy graph

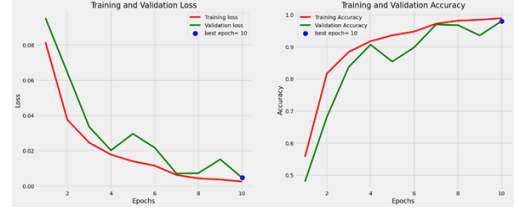


Fig. 3. Epoch vs loss/accuracy curve for model B

F. Algorithm

Following diagram, fig 4, demonstrates the process of model development implemented to create the system.

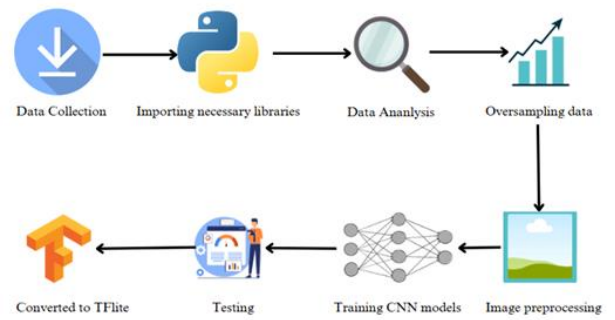


Fig. 4. Model Development process

- The HAM10000 dataset, which is a source of over 10000 dermatoscopic images, is collected from Harvard dataverse.
- Libraries like numpy, pandas, matplotlib, RandomOverSampler etc, are imported.
- The dataset is analysed to understand underlying relationships between data and different factors like localization, age and gender distribution.
- Oversampling of data is performed which increases the data samples from 10005 images to 46935 images.
- Image preprocessing techniques which include rotation, zooming, rescaling and normalization of pixels are applied.
- Data is split between training and testing counterparts in the ratio of 80 to 20. Two CNN model architectures are defined tailored to custom activation and loss function.
- The two models are tested and converted to Tflite format to then be ensembled.
- The two models are aggregated using average function to form an ensemble model. This is converted to a TFLite format for use.

Fig. 5 and fig. 6 showcase the multilayered structure of model A and model B respectively and explain the model architecture in a detailed manner.

Model: "sequential"		
Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 28, 28, 32)	896
max_pooling2d (MaxPooling2D)	(None, 14, 14, 32)	0
batch_normalization (BatchNormalization)	(None, 14, 14, 32)	128
conv2d_1 (Conv2D)	(None, 14, 14, 64)	18,496
conv2d_2 (Conv2D)	(None, 14, 14, 64)	36,928
max_pooling2d_1 (MaxPooling2D)	(None, 7, 7, 64)	0
batch_normalization_1 (BatchNormalization)	(None, 7, 7, 64)	256
conv2d_3 (Conv2D)	(None, 7, 7, 128)	73,856
conv2d_4 (Conv2D)	(None, 7, 7, 128)	147,584
max_pooling2d_2 (MaxPooling2D)	(None, 3, 3, 128)	0
batch_normalization_2 (BatchNormalization)	(None, 3, 3, 128)	512
conv2d_5 (Conv2D)	(None, 3, 3, 256)	295,168
conv2d_6 (Conv2D)	(None, 3, 3, 256)	590,880
max_pooling2d_3 (MaxPooling2D)	(None, 1, 1, 256)	0
flatten (Flatten)	(None, 256)	0
dropout (Dropout)	(None, 256)	0
dense (Dense)	(None, 256)	65,792
batch_normalization_3 (BatchNormalization)	(None, 256)	1,024
dense_1 (Dense)	(None, 128)	32,896
batch_normalization_4 (BatchNormalization)	(None, 128)	512
dense_2 (Dense)	(None, 64)	8,256
batch_normalization_5 (BatchNormalization)	(None, 64)	256
dense_3 (Dense)	(None, 32)	2,080
batch_normalization_6 (BatchNormalization)	(None, 32)	128
classifier (Dense)	(None, 7)	231

Total params: 1,275,079 (4.86 MB)
Trainable params: 1,273,671 (4.86 MB)
Non-trainable params: 1,408 (5.50 KB)

Fig. 5. Model A summary

Model: "sequential_1"		
Layer (type)	Output Shape	Param #
conv2d_7 (Conv2D)	(None, 28, 28, 32)	896
max_pooling2d_4 (MaxPooling2D)	(None, 14, 14, 32)	0
batch_normalization_7 (BatchNormalization)	(None, 14, 14, 32)	128
conv2d_8 (Conv2D)	(None, 14, 14, 64)	18,496
conv2d_9 (Conv2D)	(None, 14, 14, 64)	36,928
max_pooling2d_5 (MaxPooling2D)	(None, 7, 7, 64)	0
batch_normalization_8 (BatchNormalization)	(None, 7, 7, 64)	256
conv2d_10 (Conv2D)	(None, 7, 7, 128)	73,856
conv2d_11 (Conv2D)	(None, 7, 7, 128)	147,584
max_pooling2d_6 (MaxPooling2D)	(None, 3, 3, 128)	0
batch_normalization_9 (BatchNormalization)	(None, 3, 3, 128)	512
conv2d_12 (Conv2D)	(None, 3, 3, 256)	295,168
conv2d_13 (Conv2D)	(None, 3, 3, 256)	590,880
max_pooling2d_7 (MaxPooling2D)	(None, 1, 1, 256)	0
flatten_1 (Flatten)	(None, 256)	0
dropout_1 (Dropout)	(None, 256)	0
dense_4 (Dense)	(None, 256)	65,792
batch_normalization_10 (BatchNormalization)	(None, 256)	1,024
dense_5 (Dense)	(None, 128)	32,896
batch_normalization_11 (BatchNormalization)	(None, 128)	512
dense_6 (Dense)	(None, 64)	8,256
batch_normalization_12 (BatchNormalization)	(None, 64)	256
dense_7 (Dense)	(None, 32)	2,080
batch_normalization_13 (BatchNormalization)	(None, 32)	128
classifier (Dense)	(None, 7)	231

Total params: 1,275,079 (4.86 MB)
Trainable params: 1,273,671 (4.86 MB)
Non-trainable params: 1,408 (5.50 KB)

Fig. 6. Model B summary

V. RESULTS AND DISCUSSION

In this melanoma detection project, we achieved exemplary results with the model having noteworthy performance. The confusion matrix, fig. 7, indicates high accuracy of the ensemble model and highlights that it made 1359 correct predictions of elements of class 6 i.e. melanoma.

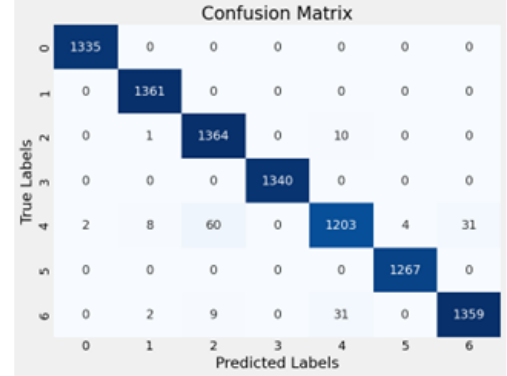


Fig. 6. Confusion matrix of ensemble model

The correlation matrix visualises the relationship between all the possible pairs between the different predictions of the two models. The negative values of the non-diagonal elements in fig. 8 represent weak negative correlations between the different predictions.

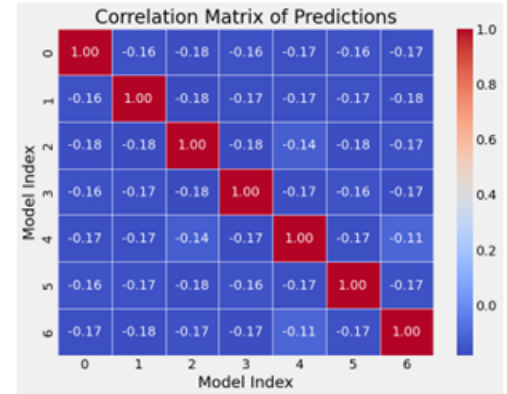


Fig. 7. Correlation matrix of ensemble model

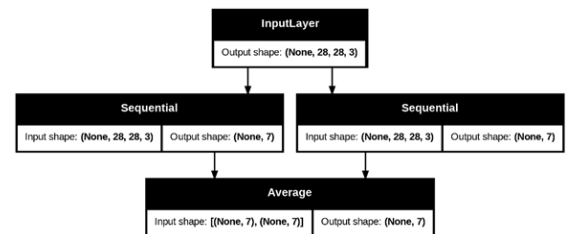


Fig. 8. Architecture of ensemble model

The ensemble model layout is explained in given fig. 9. The first component is the input layer which receives 28x28 pixel images with RGB colour channels as input. The two sequential models then separately give class predictions as

output. These outputs are combined by the average layer to provide the final result.

Given table 1 provides the full comparison of the various CNN based models on the basis of training and testing accuracy and loss. Model A and model B are CNN models with custom activation function and dense layers and uses convolutional and pooling layers for feature extraction and batch normalisation to stabilise the learning. Model A uses CCE as loss function and ReLU as activation function to achieve a test accuracy of 96.52. Model B on the other hand uses MSE as activation function and Leaky ReLU as activation function to achieve a test accuracy of 98.16.

It is found that the aggregation of the two models provide the best results with the accuracy of 98.5. As compared to the models tested with the proposed model by CK Viknesh et al[7], our proposed model attained an accuracy greater than 7.5 %.

TABLE I. COMPARISON OF MODEL PREDICTION ACCURACIES

S.no.	Models	Database	Training Accuracy	Testing accuracy
1	Model A	ISIC	97.8	96.52
2	Model B	ISIC	99.08	98.16
3	Ensemble model of model A and model B	ISIC	99.5	98.5
4	Model in [7]	ISIC	94	91

VI. CONCLUSION

Both Model A and Model B's architectural layout provides an overview of their neural network structures and the inner working of the systems that helps in the extraction of features necessary for accurate lesion classification and prediction. The convolutional layers capture the spatial hierarchies, while batch normalization ensures stable and efficient training by maintaining mean and standard deviation. Dropout layers resolve the problem of overfitting, enhancing the models' generalization capabilities. The dense classifier processes the acquired features to categorize skin lesions into predicting melanoma and non-melanoma classes, leading to a comprehensive diagnostic structure.

```
print(f"Accuracy: {accuracy:.4f}")
print(f"Precision: {precision:.4f}")
print(f"Recall: {recall:.4f}")
print(f"F1-Score: {f1:.4f}")

Accuracy: 0.9850
Precision: 0.9851
Recall: 0.9850
F1-Score: 0.9848
```

Fig. 10. Ensemble accuracy

These two models are then combined together by averaging the separate outputs of each to result in prediction of the ensemble. Through the acquired TFlite model,

predictions can be made with up to 0.9850 accuracy as seen in fig. 10.

This can further be integrated with an android mobile application to allow users to upload a picture of the lesion and predict the presence or absence of melanoma.

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REFERENCES

- [1] Autier, P. and Doré for Epimel and Eortc Melanoma Cooperative Group, J.-F. (1998), Influence of sun exposures during childhood and during adulthood on melanoma risk. *Int. J. Cancer*, 77: 533-537. [https://doi.org/10.1002/\(SICI\)1097-0215\(19980812\)77:4<533::AID-IJC10>3.0.CO;2-7](https://doi.org/10.1002/(SICI)1097-0215(19980812)77:4<533::AID-IJC10>3.0.CO;2-7)
- [2] Schadendorf, D., Fisher, D., Garbe, C. et al. Melanoma. *Nat Rev Dis Primers* 1, 15003 (2015). <https://doi.org/10.1038/nrdp.2015.3>
- [3] Wang JY, Wang EB, Swetter SM. What Is Melanoma? *JAMA*. 2023;329(11):948. doi:10.1001/jama.2022.24888. <https://jamanetwork.com/journals/jama/fullarticle/2802151>
- [4] Domingues, B., Lopes, J. M., Soares, P., & Pópulo, H. (2018). Melanoma treatment in review. *ImmunoTargets and Therapy*, 7, 35–49. <https://doi.org/10.2147/ITT.S134842>
- [5] <https://www.dana-farber.org/cancer-care/types/melanoma/diagnosis>
- [6] A. A. Adegun and S. Viriri, "Deep Learning-Based System for Automatic Melanoma Detection," in *IEEE Access*, vol. 8, pp. 7160-7172, 2020, doi: 10.1109/ACCESS.2019.2962812. <https://ieeexplore.ieee.org/abstract/document/8945133/references#references>
- [7] Viknesh, C. K., Kumar, P. N., Seetharaman, R., & Anitha, D. (2023). Detection and Classification of Melanoma Skin Cancer Using Image Processing Technique. *Diagnostics (Basel, Switzerland)*, 13(21), 3313. <https://doi.org/10.3390/diagnostics13213313>
- [8] Moturi, D., Surapaneni, R.K. & Avanigadda, V.S.G. Developing an efficient method for melanoma detection using CNN techniques. *J Egypt Natl Canc Inst* 36, 6 (2024). <https://doi.org/10.1186/s43046-024-00210-w>
- [9] Balazs Harangi, Skin lesion classification with ensembles of deep convolutional neural networks, *Journal of Biomedical Informatics*, Volume 86, 2018, Pages 25-32, ISSN 1532-0464, <https://doi.org/10.1016/j.jbi.2018.08.006>.
- [10] C. P., M. ., P. Jo, J. ., & V., V. . (2023). Computer-Aided Detection of Skin Cancer Detection from Lesion Images Via Deep Learning Techniques: 3d CNN Integrated Inception V3 Networks. *International Journal of Intelligent Systems and Applications in Engineering*, 11(9s), 550-562. Retrieved from <https://ijisae.org/index.php/IJISAE/article/view/3191>
- [11] Kumar Lilhore, U., Simaiya, S., Sharma, Y.K. et al. A precise model for skin cancer diagnosis using hybrid U-Net and improved MobileNet-V3 with hyperparameters optimization. *Sci Rep* 14, 4299 (2024). <https://doi.org/10.1038/s41598-024-54212-8>
- [12] Li Y, Shen L. Skin Lesion Analysis towards Melanoma Detection Using Deep Learning Network. *Sensors*. 2018; 18(2):556. <https://doi.org/10.3390/s18020556>
- [13] Jinen Daghrir, Lotfi Tlig, Moez Bouchouicha, Mounir Sayadi. Melanoma skin cancer detection using deep learning and classical machine learning techniques: A hybrid approach. *International Conference on Advanced Technologies for Signal and Image*

Processing, Sep 2020, Sfax, Tunisia.
 ff10.1109/ATSP49331.2020.9231544ff. fhal-03172718
<https://hal.science/hal-03172718/file/PID6365049.pdf>

- [14] Banerjee, S., Singh, S. K., Chakraborty, A., Das, A., & Bag, R. (2020). Melanoma diagnosis using deep learning and fuzzy logic. *Diagnostics*, 10(8), 577. <https://www.mdpi.com/2075-4418/10/8/577>
- [15] Alwakid, G., Gouda, W., Humayun, M., & Sama, N. U. (2022, December). Melanoma detection using deep learning-based classifications. In *Healthcare* (Vol. 10, No. 12, p. 2481). MDPI. <https://doi.org/10.3390/healthcare10122481>
- [16] Ansari, U. B., & Sarode, T. (2017). Skin cancer detection using image processing. *Int. Res. J. Eng. Technol.*, 4(4), 2875-2881. <https://www.academia.edu/download/53555135/IRJET-V4I4702.pdf>
- [17] Kousis, I., Perikos, I., Hatzilygeroudis, I., & Virvou, M. (2022). Deep learning methods for accurate skin cancer recognition and mobile application. *Electronics*, 11(9), 1294. <https://doi.org/10.3390/electronics11091294>
- [18] Dascalu, A., & David, E. O. (2019). Skin cancer detection by deep learning and sound analysis algorithms: A prospective clinical study of an elementary dermoscope. *EBioMedicine*, 43, 107-113.
- [19] Jojoa Acosta, M. F., Caballero Tovar, L. Y., Garcia-Zapirain, M. B., & Percybrooks, W. S. (2021). Melanoma diagnosis using deep learning techniques on dermatoscopic images. *BMC Medical Imaging*, 21, 1-11. <https://link.springer.com/content/pdf/10.1186/s12880-020-00534-8.pdf>
- [20] Nawaz, M., Mehmood, Z., Nazir, T., Naqvi, R. A., Rehman, A., Iqbal, M., & Saba, T. (2022). Skin cancer detection from dermoscopic images using deep learning and fuzzy k - means clustering. *Microscopy research and technique*, 85(1), 339-351. <https://doi.org/10.1002/jemt.23908>
- [21] O'Shea, Keiron & Nash, Ryan. (2015). An Introduction to Convolutional Neural Networks. ArXiv e-prints. https://www.researchgate.net/publication/285164623_An_Introduction_to_Convolutional_Neural_Networks
- [22] Sakshi Indolia, Anil Kumar Goswami, S.P. Mishra, Pooja Asopa, Conceptual Understanding of Convolutional Neural Network- A Deep Learning Approach, *Procedia Computer Science*, Volume 132, 2018, Pages 679-688, ISSN 1877-0509, <https://doi.org/10.1016/j.procs.2018.05.069>.
- [23] Yamashita, R., Nishio, M., Do, R.K.G. et al. Convolutional neural networks: an overview and application in radiology. *Insights Imaging* 9, 611–629 (2018). <https://doi.org/10.1007/s13244-018-0639-9>
- [24] Srivastava, N., Hinton, G., Krizhevsky, A., Sutskever, I., & Salakhutdinov, R. (2014). Dropout: a simple way to prevent neural networks from overfitting. *The journal of machine learning research*, 15(1), 1929-1958.
- [25] Bjorck, N., Gomes, C. P., Selman, B., & Weinberger, K. Q. (2018). Understanding batch normalization. *Advances in neural information processing systems*, 31.
- [26] Terven, Juan & Cordova-Esparza, Diana-Margarita & Ramirez-Pedraza, Alfonzo & Chávez Urbiola, Edgar. (2023). Loss Functions and Metrics in Deep Learning. A Review. 10.48550/arXiv.2307.02694.
- [27] Gordon-Rodriguez, E., Loaiza-Ganem, G., Pleiss, G., & Cunningham, J. P. (2020). Uses and abuses of the cross-entropy loss: Case studies in modern deep learning.
- [28] Shanshan Huang, Xin Jin, Qian Jiang, Li Liu, Deep learning for image colorization: Current and future prospects, *Engineering Applications of Artificial Intelligence*, Volume 114, 2022, 105006, ISSN 0952-1976, <https://doi.org/10.1016/j.engappai.2022.105006>.
- [29] Sharma, Sagar, Simone Sharma, and Anidhya Athaiya. "Activation functions in neural networks." *Towards Data Sci* 6.12 (2017): 310-316. <https://www.ijeast.com/papers/310-316,Tesma412,IJEAST.pdf>
- [30] Szandała, Tomasz. (2020). Review and Comparison of Commonly Used Activation Functions for Deep Neural Networks. 10.48550/arXiv.2010.09458.
- [31] <https://machinelearningmastery.com/random-oversampling-and-undersampling-for-imbalanced-classification/> J. Clerk Maxwell, A Treatise on Electricity and Magnetism, 3rd ed., vol. 2. Oxford: Clarendon, 1892, pp.68–73.